STACKS, DERIVED CATEGORIES OF SPACES: RESOLUTION-PROPERTY-GOES-DOWN-FINITE-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7194-7200 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 7202-7225, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 52 of queue_spaceperfect.json (pb=809, stmt 7194-7200, proof 7202-7225).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-resolution-property-goes-down-finite-flat' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DK2: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7225–7225. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a surjective finite locally free morphism of algebraic spaces over S. If X has the resolution property, so does Y.*

Proof. The condition means that f is affine and that $f_*\mathcal{O}_X$ is a finite locally free $\mathcal{O}_Y$-module of positive rank. Let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module of finite type. By assumption there exists a surjection $\mathcal{E} \to f^*\mathcal{G}$ for some finite locally free $\mathcal{O}_X$-module $\mathcal{E}$. Since f_* is exact (Cohomology of Spaces, Section 0DK2) we get a surjection

$$f_*\mathcal{E} \longrightarrow f_*f^*\mathcal{G} = \mathcal{G} \otimes_{\mathcal{O}_Y} f_*\mathcal{O}_X$$

Taking duals we get a surjection

$$f_*\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, \mathcal{O}_Y) \longrightarrow \mathcal{G}$$

Since $f_*\mathcal{E}$ is finite locally free, we conclude. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.